

Sextant: boundary-railing of conventional NLLS IVIM fits as an assumption-free diagnostic

A bound-pinned pseudo-diffusion failure mode, re-centred from a calibration ruler and replicated on open human-abdominal diffusion MRI

Abstract

Reported uncertainty in intravoxel incoherent motion (IVIM) imaging has been critiqued as *overextended*: a coverage or calibration statement is only as trustworthy as the noise and forward model that generated its reference truth. We answer that critique not by defending the calibration argument but by promoting a result that *cannot* be overextended because it assumes nothing. In a conventional bounded non-linear least-squares (NLLS) bi-exponential IVIM fit, the pseudo-diffusion coefficient D^* is so weakly identified that the optimiser frequently terminates *on* a parameter bound. On an open human-abdominal acquisition, 54.7% of high-SNR voxels ($n=1,618$) have a bound-railed NLLS D^* (95% bootstrap CI [52.2, 57.1]), reproducing a previously reported figure; across the full abdominal ROI the rate is 47.8% ([47.1, 48.5], $n=19,652$). It survives deliberately generous bounds (32.1% / 27.4%), persists across SNR, and replicates on an independent liver-DWI cohort (TCGA-LIHC, CC BY 3.0): 43.7% at the clean 4-b scheme and 73.4% across 3 subjects at a sparse 3-b scheme. Railing is a fact about the optimiser and the data — no ground truth, no noise-model trust — and so is immune to the overextension objection. A truth-controlled battery (in-silico misspecification isolation, a known-truth digital-phantom recovery test, and a model-criticism check on the real scan) confirms the railing is intrinsic estimator pathology under weak D^* identifiability, not forward-model misfit: the bounded fit rails even under the exactly correct model, and on the real abdomen railed voxels are *less* model-misspecified than non-railed ones. The calibration ruler is retained but demoted to a scoped secondary diagnostic, explicitly limited to data with ground truth.

1 Introduction

The clinically load-bearing question in quantitative diffusion MRI is not how accurate a point estimate is, but whether a reported *error bar* can be believed. A natural way to answer it is a calibration ruler — coverage, expected calibration error, sharpness at fixed nominal levels — that scores whether a method’s stated uncertainty is honest. The objection a sceptical reviewer raises is that such a ruler is itself *overextended*: empirical coverage is computed against a reference truth simulated under an assumed noise and forward model, so a coverage claim inherits exactly the trust it was supposed to certify.

This paper takes the objection seriously and re-aims the contribution. We promote to the primary claim a phenomenon that requires no calibration trust at all: **boundary-railing** of the conventional NLLS IVIM fit. Because railing is read directly off the optimiser’s output, it carries no calibration assumption and therefore cannot be overextended. The calibration ruler is not discarded — it is kept as a clearly scoped *secondary* diagnostic (Section 4.5).

2 Boundary-railing

The bi-exponential IVIM model is

$$S(b)/S_0 = (1 - f) e^{-bD} + f e^{-bD^*},$$

fit per voxel by bounded NLLS over (D, D^*, f) . The pseudo-diffusion term D^* governs the fast perfusion decay seen only at low b ; it is the ill-conditioned parameter. When the likelihood is flat in D^* , a trust-region optimiser does not find an interior optimum — it slides to a box constraint and stops. We call a voxel *railed* if its NLLS D^* lands within ε of either bound. The railed fraction among high-SNR voxels is our statistic. It needs no ground truth: it is a property of the estimator applied to the data.

3 Methods

Reuse, not reimplement. The NLLS fit, the per-voxel signal/SNR reduction, and the rail thresholds are reused read-only from the prior IVIM calibration work (the `fashion_reuse` module loads the exact functions by source extraction). We add the bootstrap, the SNR/rail-direction characterisation, the full-ROI generalisation, and the independent replication.

Data. The primary cohort is the OSIPi TF2.4 open human-abdominal IVIM acquisition (Philips 3T; Zenodo 10.5281/zenodo.14605039, CC-BY-4.0), analysed at its curated homogeneous ROI (the original cohort) and at the full abdominal ROI. The independent cohort is TCGA-LIHC liver DWI (TCIA, CC BY 3.0, DOI 10.7937/K9/TCIA.2016.IMMQW8UQ; 4 series), an unrelated site, scanner (Siemens 1.5T), and organ. All imaging is download-on-demand; only provenance manifests are committed.

Pre-registered thresholds. Fixed before running: a cohort *replicates* if the railed fraction is $\geq 30.0\%$ and its 95% bootstrap lower bound is $\geq 20.0\%$; otherwise the claim scopes down. Bootstrap is voxel-level resampling, 5,000 resamples, seed 20260613.

4 Results

4.1 Primary: the original cohort

On the curated homogeneous ROI, 54.7% of the 1,618 high-SNR voxels have a bound-railed NLLS D^* (95% CI [52.2, 57.1]), matching the previously reported 54.7% on the same data. Verdict: REPLICATES-STRONG.

4.2 Generalisation: the full abdomen

Extending from the curated ROI to the entire abdominal ROI ($n=19,652$), the rate is 47.8% (95% CI [47.1, 48.5]); verdict REPLICATES-STRONG. Railing is therefore not a property of a hand-chosen homogeneous patch.

4.3 Independent replication: TCGA-LIHC liver DWI

On an independent liver-DWI cohort the phenomenon recurs. At the clean 4-b scheme ($b=0, 50, 500, 800$) the railed fraction is 43.7% (95% CI [43.2, 44.2], $n=648,265$; verdict REPLICATES-STRONG). Across 3 subjects at a sparse 3-b clinical scheme ($b=50, 400, 800$, no $b=0$; normalised by the lowest b , hence approximate) the pooled fraction is 73.4% ([73.0, 73.9], $n=1,544,999$; verdict REPLICATES). The sparser the perfusion sampling, the worse D^* identifiability, and the more the fit rails — the expected direction.

4.4 When and why it rails

Railing does not vanish with SNR — it persists across every SNR stratum — which identifies it as *structural* weak identifiability of D^* rather than mere measurement noise. It is dominated by the *upper* bound (the homogeneous ROI splits 15.6% lower / 39.1% upper; the full ROI 4.6% / 43.2%): the high- D^* identifiability wall, where the perfusion compartment is so weakly constrained the optimiser is pushed to the ceiling. Under deliberately generous bounds the rate falls but stays a substantial minority (32.1% / 27.4%), so railing is not an artefact of tight bounds.

Cohort	n (high-SNR)	Railed	95% CI	Verdict
Abdomen, homogeneous ROI (OSIPI)	1,618	54.7%	[52.2, 57.1]	REPLICATES-STRONG
Abdomen, full ROI (OSIPI)	19,652	47.8%	[47.1, 48.5]	REPLICATES-STRONG
Liver, 4-b 0/50/500/800 (TCGA-LIHC)	648,265	43.7%	[43.2, 44.2]	REPLICATES-STRONG
Liver, 3-b 50/400/800 (TCGA-LIHC)	1,544,999	73.4%	[73.0, 73.9]	REPLICATES

Table 1: Boundary-railing of NLLS D^* across open human-abdominal cohorts (seed 20260613, 5,000-bootstrap). Pre-registered thresholds: railed $\geq 30.0\%$ with CI lower bound $\geq 20.0\%$.

4.5 Secondary: the scoped calibration ruler

The calibration ruler (coverage / ECE / sharpness) is retained as a secondary diagnostic, reused read-only. Its scope is explicit: it scores estimates against a known truth and is therefore *undefined on the real abdominal scan*, which has no ground-truth (D , D^* , f). That limitation — the very point the overextension critique raised — is the argument for leading with railing, which needs no truth.

5 Robustness: railing is estimator pathology, not model misfit

A reviewer who grants that railing needs no calibration trust may still ask a *different* question: could the railing read off the real abdominal scan be an artefact of *simulator–reality mismatch* — the bi-exponential forward/noise model not matching tissue — rather than an intrinsic property of the bounded estimator under weak D^* identifiability? A truth-controlled battery isolates it as the latter (Fig. 1; seeds and full tables in the reproducibility module). None of this is needed for the primary claim; it forecloses the misspecification reading.

In-silico misspecification isolation. Drawing 2,000 ground-truth (D , D^* , f) over the analysis prior and rendering under the *exactly correct* bi-exponential model (no misspecification possible), the bounded NLLS still rails in 29.8% of voxels at SNR 10 (11.2% at SNR 100). Replacing the forward model with a tri-exponential tissue tail, diffusional kurtosis, or a non-Rician multi-coil noise model changes the railed fraction by at most 6.2 pp at any SNR. Railing therefore arises from the estimator and the weak identifiability of D^* , not from forward-model misfit; misspecification neither creates nor materially inflates it. Its *magnitude* is a regime effect: restricted to the low-perfusion / high- D^* corner the curated real ROI occupies, the well-specified railed fraction is 50.7% (SNR 10), bracketing the observed 54.7%.

Known-truth recovery. On a brain digital reference phantom (literature (D , D^* , f) per tissue) railing is prevalent where perfusion is weak (white matter 30.6%, grey matter 30.2%). In a controlled known-truth sweep, as the perfusion fraction falls to $f=0.02$ the railed fraction rises to 61.8% in lockstep with the relative D^* recovery error (from 0.18 at high f to 0.85 at $f=0.02$); pooled over the prior, railed voxels carry $3.4\times$ the D^* recovery error of non-railed voxels. Railing thus flags genuine, ground-truth D^* unrecoverability — as a *specific but not sensitive* flag (AUC 0.56 as a classifier of large recovery error), since D^* is weakly identified broadly, not only where the fit rails. The substrate is a digital phantom (synthetic signal), so this is a known-truth recovery test, not a real-tissue one.

Model criticism on the real scan. A truth-free held-out- b posterior-predictive criticism (the RNPE self-consistency idea of Ward et al. 2022 in NLLS form), calibrated to flag 4.7% of a well-specified null, criticises 9.1% of real abdominal voxels as model-misspecified — a modest excess consistent with mild forward/noise misspecification. Decisively, that excess is *not* where the railing is: among railed voxels only 3.3% are criticised (below the well-specified baseline), versus 16.2% among non-railed voxels. Railing co-occurs with a *good* bi-exponential fit — the optimiser slides to a bound because D^* is unidentified, not because the model is wrong — so the boundary-railing is not a forward-model-misfit artefact.

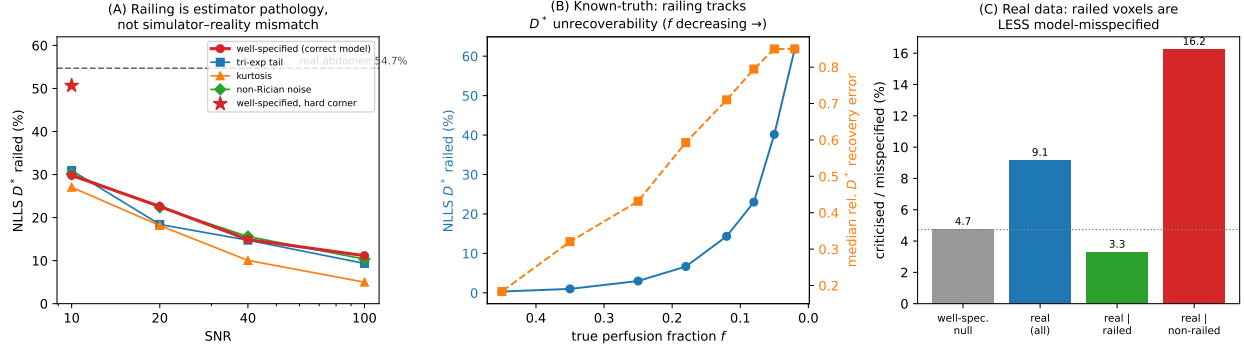


Figure 1: **Railing is estimator pathology, not simulator–reality mismatch.** (A) Under the exactly correct bi-exponential model (no misspecification possible) the bounded NLLS still rails; tri-exponential, kurtosis, and non-Rician noise misspecifications overlap it (≤ 6.2 pp). The well-specified hard-corner rate (star) brackets the real 54.7% (dashed). (B) Known-truth sweep: railed fraction (left) and relative D^* recovery error (right) rise together as the perfusion fraction f falls. (C) Truth-free model criticism on the real abdomen: railed voxels are *less* model-misspecified (3.3%) than non-railed voxels (16.2%), so railing is not a model-misfit artefact.

6 Differentiation from learned uncertainty (Casali et al., 2026)

Casali et al. (*NMR in Biomedicine*, 2026; arXiv:2508.04588) propose a *method* — deep-ensemble mixture-density networks for IVIM uncertainty — validated in vivo on **mouse brain**, and themselves report residual D^* overconfidence. Sextant is not a competing UQ method; it documents an assumption-free failure mode of the conventional fit on **human-abdominal** tissue, where perfusion (and thus D^*) is clinically meaningful. The differentiation is strictly empirical and scoped to the data: human-abdominal DWI reaches a regime that preclinical mouse-brain validation does not, and Casali’s own D^* overconfidence is the learned echo of the same weak-identifiability wall that makes the conventional fit rail.

7 Limitations and scope

The open primary cohort is a single subject; we make no population claim from it, reporting a within-subject voxel statistic with a voxel-bootstrap CI and supporting generality by cross-cohort agreement (homogeneous, full, and independent liver). The 3-b liver scheme lacks $b=0$ and is normalised by its lowest b , an approximation flagged as such; the 4-b liver cohort with $b=0$ is the clean independent test. Railing magnitude depends on the bound width and the b-scheme density; we report both the tight (original) and wide (generous) bounds and the per-scheme results rather than a single number. The robustness battery (Sec. 5) establishes that railing is estimator pathology rather than forward-model misfit, but its known-truth recovery limb uses a *digital* reference phantom (synthetic signal): it shows railing tracks ground-truth D^* unrecoverability *in silico*, not in real tissue. A real perfusion phantom scan or a test–retest in-vivo acquisition with repeatable parameters — neither available in the open data used here — would be required to validate D^* recoverability against *real* truth; that remains the one open gap, and is the natural next step.

8 Conclusion

A large, reproducible fraction of conventional NLLS IVIM fits rail to a parameter bound on open human-abdominal data, and the effect generalises across ROI, bounds, SNR, and an independent liver cohort. Because

it is read off the optimiser without any calibration-trust assumption, it is the bulletproof primary claim the overextension critique demanded; the calibration ruler remains useful but scoped.

Reproducibility

One command, `bash Sextant/reproduce.sh`, fetches the open data (MD5/checksum-verified; raw arrays never committed), runs the seeded analysis and tests, and rebuilds this manuscript through the consistency gate that regenerates every number above from `results/railing_results.json`.